MAT2001 Statistics for Engineers L T P J C 3 0 0 0 3

Pre-requisite: Calculus and Matrices

Course Educational Objectives

The course aims to:
• Provide the basic concepts of probability and random variables.
• Introduce sampling distributions and estimation.
• Teach hypothesis testing and regression analysis.

Course Learning Outcomes

Upon successful completion of the course, students will be able to:
CO1: Compute probabilities and expectations of random variables.
CO2: Estimate population parameters using sampling distributions.
CO3: Perform tests of hypotheses and fit regression models.

UNIT I: Probability and Random Variables (9 hours)
Axioms of probability, conditional probability, Bayes theorem, discrete and continuous random variables, expectation, variance, moment generating functions.

UNIT II: Sampling and Estimation (9 hours)
Sampling distributions, central limit theorem, point estimation, confidence intervals.

UNIT III: Testing of Hypotheses (9 hours)
Large sample tests, t-test, chi-square test, F-test, analysis of variance.

UNIT IV: Correlation and Regression (9 hours)
Scatter diagrams, correlation coefficient, simple linear regression, multiple regression, time series.

Total Hours: 36

Text Books

1. Ronald E. Walpole, Probability and Statistics for Engineers and Scientists, 9th Edition, Pearson, 2012.
2. Jay L. Devore, Probability and Statistics for Engineering and the Sciences, 9th Edition, Cengage, 2016.

Reference Books

1. Sheldon Ross, A First Course in Probability, 9th Edition, Pearson, 2014.